Repeated `tc` error messages in network attacks

BusinessMap ticket **12109** · Fix report · 7 August 2026 · Antoine Choimet · steadybit/action-kit PR [#476]

> **Outcome.** When a network attack's `tc` batch failed, the same error line was shown once per failed rule — producing a wall of identical text in the action's error details. Identical messages are now collapsed to a single occurrence with an `(and N more)` counter. Verified end to end with the real extension: the error details for a 6-port delay attack shrank from **257 lines to 8**, with no information lost. Shipped in `action_kit_commons v1.10.5`.

What the customer saw

Running a network delay attack on a custom Ubuntu image (Zalando), the action failed and the error details repeated the same two-line kernel error dozens of times, once per `tc` rule the attack tried to install.

Root cause

`tc` is invoked in batch mode (`tc -force -batch -`), and it reports one error per failed batch line. A network attack installs one rule per interface, address family, direction and port — so a single delay attack easily submits dozens of lines. When an early line fails (typically the root qdisc), every subsequent line fails with the *same* message.

The renderer for these batch errors printed every entry verbatim, so the user-facing message grew linearly with the number of rules, even though it only ever contained two or three distinct errors.

The fix

In `go/action_kit_commons/network/netfault/batch_error.go`, the rendering of `batchErrors` now deduplicates identical messages: the first occurrence is kept — with its `Command failed -:<line>` reference — and repeats collapse into an `(and N more)` suffix.

- Only the *rendering* changed. The parsed error list is untouched, so the ignore-filtering of benign cleanup errors and the kernel-configuration hints behave exactly as before.
- No caller parses the rendered string, so nothing downstream is affected.
- Reviewed and merged as PR #476; an automated review found no correctness, security or performance issues.

Proof: end-to-end run with the real extension

Both runs below used the **actual extension-host binary** in a privileged container on a Linux kernel (6.12.76), driven through its HTTP action API exactly as the platform drives it (`POST /prepare` then `POST /start`). Nothing was simulated: real `tc`, real kernel errors, real action response.

Setup, identical for both runs

- **Attack:** network delay, 500 ms, on 6 ports — which expands to **86 `tc` batch lines**, comparable to the customer's multi-rule attack.

- **Failure trigger:** the target interface was given a pre-existing root qdisc occupying handle `1:` (`pfifo_fast 1:`). The attack's first batch line, `qdisc replace dev … root handle 1: prio`, then fails, and every following filter line fails identically — the same cause-and-effect as the ticket.
- **Isolation:** the attack targeted a dedicated dummy interface, so no real network traffic was affected.
- **Only variable:** the library version — two locally built images differing solely in `action_kit_commons`.

BEFORE action_kit_commons v1.10.4

```
ERROR TITLE : Failed to start action.
ERROR DETAIL:
Failed to apply network settings.: Command failed tc -force -batch -
Error: Invalid qdisc name.
Command failed -:1
Error: Parent qdisc is not classful.
Command failed -:2
Error: Qdisc not classful.
We have an error talking to the kernel
Command failed -:3
Error: Qdisc not classful.
We have an error talking to the kernel
Command failed -:4
Error: Qdisc not classful.
We have an error talking to the kernel
Command failed -:5
                    ⋮
    the same two lines repeat for every rule,
    all the way through  Command failed -:86
                    ⋮
Error: Qdisc not classful.
We have an error talking to the kernel
Command failed -:86
```

257 lines in total. Abridged here for the page; the untruncated transcript is on file.

AFTER action_kit_commons v1.10.5

```
ERROR TITLE : Failed to start action.
ERROR DETAIL:
Failed to apply network settings.: Command failed tc -force -batch -
Error: Invalid qdisc name.
Command failed -:1
Error: Parent qdisc is not classful.
Command failed -:2
Error: Qdisc not classful.
We have an error talking to the kernel
Command failed -:3 (and 83 more)
```

Complete, unabridged output — this is the entire message.

Measured

Metric	Before (v1.10.4)	After (v1.10.5)
Lines in the error details	257	8
Occurrences of `Qdisc not classful`	84	1
Distinct errors reported	3	3
Failed `tc` lines the operator can still count	86	86

The last two rows are the important ones: nothing was hidden. All three distinct errors are still reported, the first failing line number is still shown, and the repeat count still tells the operator how many rules were affected.

Regression coverage

- A test replaying the ticket's exact `tc` stderr — including the customer's `NLM_F_REPLACE needed to override` line — asserts the collapsed output. It fails if the deduplication is ever removed.
- It runs in CI on every commit as part of the `netfault` package suite (`go-action-kit-commons` job), on Linux and Windows.

> **One caveat, for accuracy.** The customer's exact wording was `Error: Parent Qdisc doesn't exists.`, which comes from the older `tc qdisc add` behaviour in the `action_kit_commons v1.2.17` their extension was running. Current versions use `tc qdisc replace`, so the specific kernel message differs. The failure *shape* — one failing root-qdisc line followed by N identical repeats — and the fix are the same, which is what the run above demonstrates.

Rollout

Component	Version	State
action_kit_commons	v1.10.5	Merged and released (PR #476)
extension-container	v1.7.6	Dependency bumped, pending release
extension-host	v1.7.4	Dependency bumped, pending release

Customers get the improved message by upgrading either extension to the versions above; no configuration change is required.